STAT 24620 / FINM 34700 / STAT 32950
Homework 3

Due: 05/10/2026 11:59pm

Problem 1: Predictor-side dimension reduction versus supervised prediction

Suppose $Y \in \mathbb{R}^{n\times q}$ and $X \in \mathbb{R}^{n\times p}$ are both column-centered, and consider the multivariate linear model

$$Y = XB + E, \qquad B \in \mathbb{R}^{p\times q}.$$

Assume throughout this problem that the predictor variables have already been rotated into an orthogonal coordinate system, so that

$$X^\top X = \operatorname{diag}(d_1^2, \dots, d_p^2), \qquad d_1 \ge d_2 \ge \cdots \ge d_p > 0.$$

In this coordinate system, the first predictor direction has the largest sample variation, the second has the next largest, and so on. Let $\widehat{B}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top Y$, and write $\hat{b}_j^\top$ and $b_j^\top$ for the jth rows of $\widehat{B}_{\mathrm{OLS}}$ and B, respectively.

(a) Show that for any $B \in \mathbb{R}^{p\times q}$,

$$\|Y - XB\|_F^2 - \|Y - X\widehat{B}_{\mathrm{OLS}}\|_F^2 = \|X(\widehat{B}_{\mathrm{OLS}} - B)\|_F^2.$$

In words, what does this identity say about how much worse B fits than $\widehat{B}_{\mathrm{OLS}}$?

(b) Use the diagonal form of $X^\top X$ to show that

$$\|X(\widehat{B}_{\mathrm{OLS}} - B)\|_F^2 = \sum_{j=1}^{p} d_j^2 \|\hat{b}_j - b_j\|_2^2.$$

Interpret this formula. Which predictor directions are more costly to approximate poorly?

(c) Principal components regression (PCR) first keeps the predictor directions with the largest sample variation, and then regresses Y on those retained directions. Under the present assumption $X^\top X = \operatorname{diag}(d_1^2, \dots, d_p^2)$, which predictor directions would one-dimensional PCR and two-dimensional PCR keep?

(d) Explain why the criterion in part (c) depends only on the variation structure of X, whereas the identity in part (b) shows that approximation error for predicting Y depends on both d_j^2 and the row vectors $\hat{b}_j^\top$. In particular, why can a predictor direction with relatively small variance still be important for prediction?

(e) Consider the case $p = 2$, $q = 2$, with

$$d_1^2 \gg d_2^2, \qquad \|\hat{b}_1\|_2 \approx 0, \qquad \|\hat{b}_2\|_2 \text{ is large.}$$

If you are allowed to keep only one predictor direction, which direction would one-dimensional PCR keep? Why can that choice perform poorly for predicting Y?

(f) In 4–6 sentences, compare the following two strategies:

(i) reduce the dimension of X first by PCA, then regress Y on those retained components;

(ii) use a supervised low-dimensional model for prediction.

Your answer should make clear which parts of the data each strategy uses when deciding what low-dimensional structure to keep, and why the two strategies need not retain the same directions.

Problem 2: LDA, logistic regression, and classification thresholds

Assume $Y \in \{0, 1\}$, with prior probabilities

$$\Pr(Y = 1) = \pi_1, \qquad \Pr(Y = 0) = \pi_0,$$

and suppose

$$X \mid Y = k \sim N(\mu_k, \Sigma), \qquad k = 0, 1,$$

where the two classes share the same covariance matrix Σ.

(a) Show that under the LDA model above, the conditional class probability $\Pr(Y = 1 \mid X = x)$ has the logistic-regression form

$$\Pr(Y = 1 \mid X = x) = \frac{\exp(\beta_0 + x^\top \beta)}{1 + \exp(\beta_0 + x^\top \beta)}.$$

Equivalently, show that the log-posterior odds can be written in the linear form

$$\log \frac{\Pr(Y = 1 \mid X = x)}{\Pr(Y = 0 \mid X = x)} = \beta_0 + x^\top \beta.$$

Derive explicit expressions for β_0 and β in terms of $\mu_0, \mu_1, \Sigma, \pi_0, \pi_1$.

(b) Part (a) shows that under the LDA assumptions, the induced conditional model for $Y \mid X = x$ has the same logistic form as in logistic regression. In 2–4 sentences, explain why this does *not* mean that LDA and logistic regression are the same fitted method. Your answer should explicitly mention:

(i) what LDA models first,

(ii) what logistic regression models first,

(iii) why the fitted probabilities from the two methods need not be numerically identical in finite samples.

(c) Suppose the misclassification costs are $c(1 \mid 0)$ and $c(0 \mid 1)$. Show that the Bayes rule classifies x to class 1 whenever

$$\log \frac{\Pr(Y=1 \mid X=x)}{\Pr(Y=0 \mid X=x)} > \log \frac{c(1 \mid 0)}{c(0 \mid 1)}.$$

Explain which part of the linear rule is affected by changing the costs.

(d) Use the `Pima.tr` and `Pima.te` data from the `MASS` package, as in Lecture 9. For this problem, use only the predictors `glu`, `bmi`, and `age`. Fit an LDA model on `Pima.tr`. Extract from the fitted LDA object the estimated class means, pooled covariance matrix, and class priors. Use your formulas from part (a) to construct

$$\hat{\beta}_0^{(\mathrm{LDA})} \quad \text{and} \quad \hat{\beta}^{(\mathrm{LDA})},$$

then compute the probabilities

$$\widetilde{p}_{\mathrm{LDA}}(x) = \frac{\exp(\hat{\beta}_0^{(\mathrm{LDA})} + x^\top \hat{\beta}^{(\mathrm{LDA})})}{1 + \exp(\hat{\beta}_0^{(\mathrm{LDA})} + x^\top \hat{\beta}^{(\mathrm{LDA})})}$$

on `Pima.te`. Compare $\widetilde{p}_{\mathrm{LDA}}(x)$ with the posterior probabilities returned by the fitted LDA model. Verify numerically that they agree up to numerical error.

(e) Fit a logistic regression model on `Pima.tr` using the same three predictors. On `Pima.te`, produce one scatterplot of

$$\widehat{p}_{\mathrm{logit}}(x) \quad \text{versus} \quad \widetilde{p}_{\mathrm{LDA}}(x),$$

and briefly comment on whether the two fitted conditional probabilities appear similar overall.

Problem 3: Classification on the `birthwt` dataset

Use the file `birthwt.csv`, which contains the `birthwt` data from the `MASS` package. The variable `low` indicates whether birth weight is below 2.5kg. Treat `low` as the binary outcome, and use the remaining maternal/background variables as predictors, *excluding* `bwt` itself.

Use one random 70/30 train/test split, with a fixed seed that you report.

(a) Briefly describe the predictors in the dataset. Report the training-set sample size, the test-set sample size, and the class proportions in both sets.

(b) Fit a logistic regression model on the training set, and briefly identify which predictors appear most important.

(c) Fit an LDA classifier on the same training set. Report the estimated class priors.

(d) For both methods, compute predicted probabilities on the test set. Produce one figure that compares the two ROC curves on the same axes.

(e) The area under the ROC curve (AUC) is the area under the curve obtained by plotting the true positive rate against the false positive rate over all classification thresholds. Report the AUC for both methods; for example, you may use the `auc()` function in the `pROC` package. Briefly explain what a larger AUC means in this context.

(f) Using threshold 0.5, report the confusion matrix, test accuracy, sensitivity, and specificity for each method.

(g) For the logistic regression fit, choose an alternative classification threshold by maximizing Youden's index

$$\text{sensitivity} + \text{specificity} - 1$$

on the training set only. Apply that threshold to the test set and compare the resulting confusion matrix with the threshold-0.5 version.

Problem 4: Regularized prediction on `College`

Use the file `College.csv`, which contains the `College` data from the `ISLR2` package. Let `Apps` be the response.

Construct a design matrix as follows:

- keep all main effects;
- include all pairwise interactions among the *quantitative* predictors;
- keep the categorical variable `Private` only as a main effect.

Use one random 80/20 train/test split, with a fixed seed that you report.

(a) Report the final sample size n and the number of columns p in your design matrix (excluding the intercept). Briefly comment on why this is a setting where ordinary least squares may become unstable.

(b) Fit the following methods on the training set:

(i) OLS,

(ii) ridge regression,

(iii) lasso,

(iv) elastic net.

Use cross-validation to choose the tuning parameter(s) for the regularized methods. For elastic net, the mixing parameter is denoted by α, where $\alpha = 0$ gives ridge and $\alpha = 1$ gives lasso. For this problem, try the grid

$$\alpha \in \{0.25, 0.5, 0.75\},$$

and state clearly how you selected the final elastic-net model.

(c) Report the test MSE for all four methods in one table. Which method performs best on your split? Is the gap versus OLS practically large or fairly modest?

(d) For ridge and lasso, produce coefficient-path plots and briefly describe how the paths differ.

(e) For lasso, compare the models selected by `lambda.min` and `lambda.1se`. Report the test MSE and the number of nonzero coefficients for both choices. Is the extra predictive accuracy of `lambda.min` worth the added model complexity on your split?

(f) For lasso and elastic net, report the number of nonzero coefficients in the selected model. Identify several selected interaction terms and comment on whether they seem scientifically reasonable or mainly predictive in nature.

Problem 5: High-dimensional PCA and covariance shrinkage

This problem is conceptual and analytical. No coding is required.

(a) Let S be the sample covariance matrix and consider the shrinkage estimator

$$S_\alpha = (1-\alpha)S + \alpha\tau I_p, \qquad \tau = \mathrm{tr}(S)/p, \qquad 0 \le \alpha \le 1.$$

If $S = V\mathrm{diag}(\hat\lambda_1, \dots, \hat\lambda_p)V^\top$, show that

$$S_\alpha = V\mathrm{diag}\Big((1-\alpha)\hat\lambda_1 + \alpha\bar\lambda, \dots, (1-\alpha)\hat\lambda_p + \alpha\bar\lambda\Big)\, V^\top,$$

where

$$\bar\lambda = \frac{1}{p}\sum_{j=1}^{p}\hat\lambda_j.$$

What does this imply about the eigenvectors of S_α?

(b) Let

$$\hat\lambda_j^{(\alpha)} = (1-\alpha)\hat\lambda_j + \alpha\bar\lambda$$

denote the jth eigenvalue of S_α. Show that

$$\hat\lambda_j^{(\alpha)} - \bar\lambda = (1-\alpha)(\hat\lambda_j - \bar\lambda).$$

Deduce that

$$\sum_{j=1}^{p}\left(\hat\lambda_j^{(\alpha)} - \bar\lambda\right)^2 = (1-\alpha)^2\sum_{j=1}^{p}(\hat\lambda_j - \bar\lambda)^2.$$

Briefly interpret this identity.

(c) Based on parts (a) and (b), explain why shrinkage toward $\bar\lambda I_p$ can stabilize the eigenvalue spectrum without by itself solving the interpretability problem that sparse PCA is trying to address.

(d) Keeping parts (a)–(c) in mind, a student says: "If the true leading principal component is sparse, then sparse PCA must be better than ordinary PCA." Give two reasons this statement is too strong.